INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, April - May, 2026

Degree: **B.C.A.** Semester: **Second**

Stream: **B.C.A.**

Paper Code: **BCACC201**

Paper Name: **Data Structures**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1.	Explain dynamic memory allocation.	2
2.	What do you mean by nested loops?	2
3.	Define hashing.	2
4.	What do you mean by a doubly linked list?	2
5.	What is the index of the first element in a C array?	2
6.	What is a primary concern when choosing a hash function for collision resolution by chaining?	2
7.	How does the load factor impact the performance of open addressing in hash tables?	2
8.	Write the sequence of nodes visited during the In-order traversal of a Binary Tree.	2
9.	How do you find the largest and smallest elements in an array?	2
10.	How do you access an element of a 2D array in C?	2
11.	What happens if you access an array element out of its bounds?	2

12. What is the time complexity of searching an element in a Binary Search Tree (BST) in the worst case? 2

13. What is the main advantage of using a stack to implement a non-recursive traversal of a Binary Tree? 2

14. What factors should be considered when choosing a hash function for a hash table? 2

15. How does the time complexity of building a Sparse Table compare to that of querying it? 2

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Give the demerits of a hashing index. 5

17. Differentiate between the linear and binary search. 5

18. What are the different methods of hashing? 5

19. What do you mean by collision in the context of hashing? Explain it with an example. 5

20. Prove that the total number of edges in a complete binary tree with n terminal nodes are 2(n-1). 5

21. Articulate the difference between the binary tree and binary search tree. 5

22. How the merge subroutine of the merge sort works? Explain it with an example. 5

23. What is a circular linked list? Explain it with an example. How it is different from an ordinary linked list? 5

24. What is articulation point? Explain it with an example. 5

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. How the concept of binary search tree can be used to sort n integers? Write down the pseudocode and explain it with an example. 10

26. What are inverted files? Why inverted files are used for large textual databases? **10**

27. Here is an array of ten integers: 5, 3, 8, 9, 1, 7, 0, 2, 6, 4. Suppose we partition this array using quick sort's partition function and using 5 for the pivot. Draw the resulting array after the partition finishes. Show all the intermediate steps. **10**

28. Consider the following hash function: **10**
$h(x) = x \bmod 8$.
The following set of elements are inserted to an empty hash table:
26, 34, 108, 29, 44, 67, 78, 91, 23, 81.
Use quadratic probing as the collision resolution technique.

29. Write down the pseudocode for the insertion sort algorithm. Analyze its best, worst, and average case time complexity. **10**

30. Apply the selection sort algorithm on the following list of elements to sort in ascending order: 56, 57, 92, 38, 44, 90, 61, 73. Show all the intermediate steps. **10**

31. What do you mean by best case, average case, and worst case analysis of an algorithm? Explain each one of them in the context of linear search. **10**

32. Distinguish between the following: **10**
(a) Internal search and external search.
(b) Primary key and secondary key.
(c) Internal key and external key.

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